

SOC INCIDENT REPORT

SSH Brute-Force Detection, Active Response & Jira Integration

Client / Organization:	Internal Corporate Network
Prepared By:	Md. Kawsar Hossain
Role:	SOC Analyst
Report Date:	5/9/2026
Jira Incident Ticket Ref:	KAN 98
Report Reference No.:	IR-2026-09-005
Classification:	TLP: AMBER (Confidential)

Contents

1. Executive Summary
2. Attack Summary
3. Screenshot Evidence / Alerts (SIEM)
4. Incident Ticket
5. Timeline of Detected Events
6. Source of Attack (IP Addresses & Methods Used)
7. Recommendations for Detection & Blocking

1. Executive Summary

This report details the successful detection, automated mitigation, and incident ticketing of an SSH brute-force attack against an enterprise Linux environment. During the event, an external threat actor (IP: 192.168.0.100) targeted an Ubuntu server (IP: 192.168.0.109). The anomaly was swiftly detected by the Wazuh SIEM Manager (IP: 192.168.0.111) using Custom Rule 5720 (Level 12), which identified high-velocity sshd authentication failures. Upon detection, Wazuh's Active Response engine immediately deployed a firewall-drop script, dynamically updating the host's iptables to drop all active and incoming malicious connections. Simultaneously, a SOAR webhook integration instantly generated an actionable incident ticket in Atlassian Jira (KAN-98), mapping the threat to MITRE ATT&CK T1110 (Brute Force). This end-to-end automation validates the system's capability to deliver zero-touch threat containment and streamlined SOC documentation.

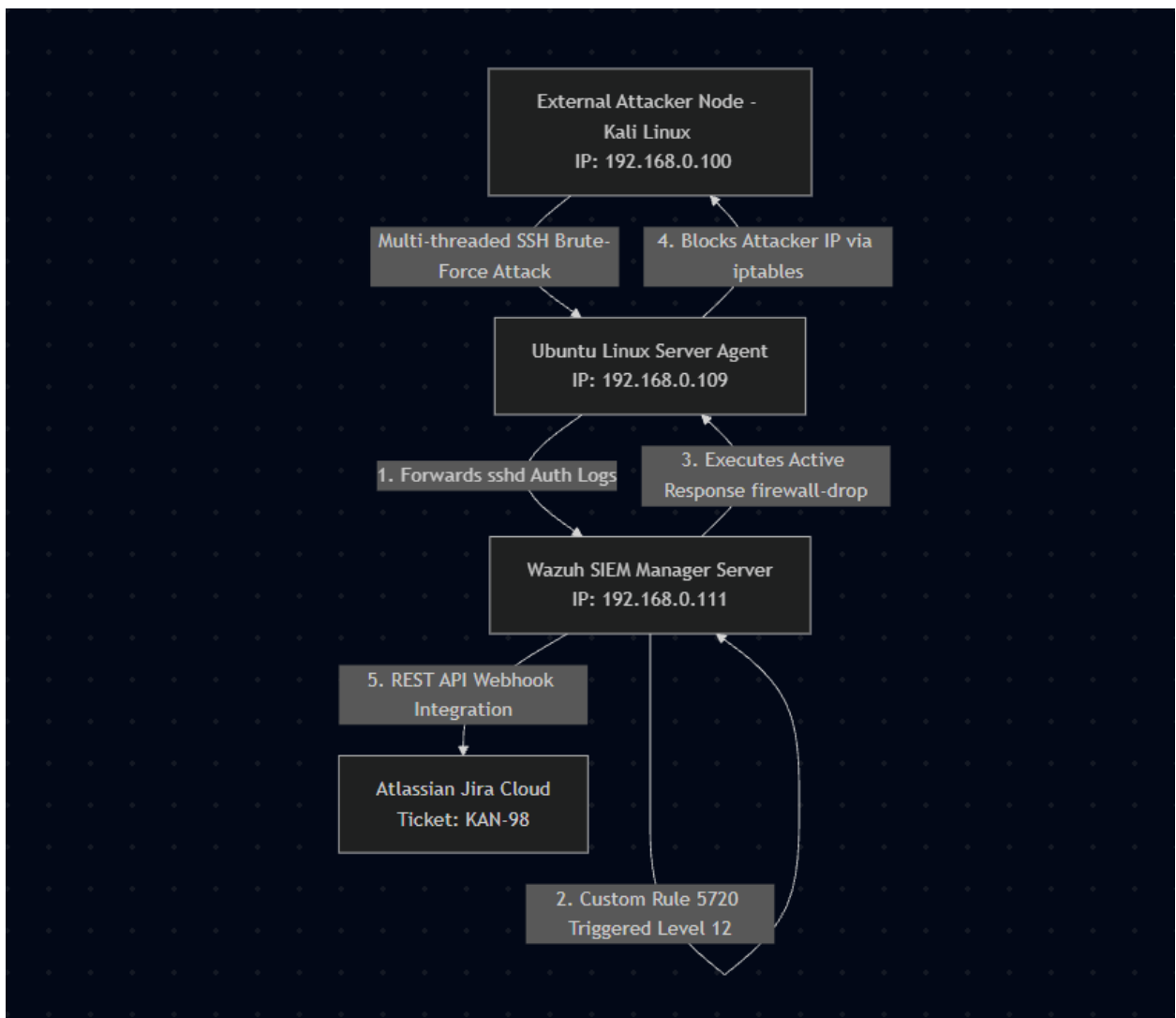
2. Attack Summary

2.1 Network / Environment Design

Target Endpoint: Ubuntu Linux Server (IP: 192.168.0.109) running sshd

SIEM/XDR Server: Production Wazuh Manager (Ubuntu)

Attacker Node: External Kali Linux Machine executing multi-threaded dictionary attacks



2.2 Tools & Systems Used

- **SIEM Platform:** Wazuh (Manager & Agent)
- **Telemetry & Logging:** Linux System Authentication Logs (sshd via system journal/syslog)
- **Remediation:** Wazuh Active Response (firewall-drop) and Jira REST API Integration

2.3 Setup & Configuration Steps

- Tuned baseline Linux authentication log parsing to filter out high-frequency noise while retaining critical failure sequences.
- Overrode Wazuh default Rule 5720 in local_rules.xml to flag high-frequency sliding-window failures (**6 failed authentication attempts within 60 seconds** from a singular source IP via same_source_ip).
- Configured Active Response (firewall-drop) inside ossec.conf linked to Rule 5720 with a temporary 120-second isolation window.
- Verified Jira integration endpoints (Rule level 10+) to ensure automatic ticket generation upon threat escalation.

2.4 Reconnaissance / Scanning

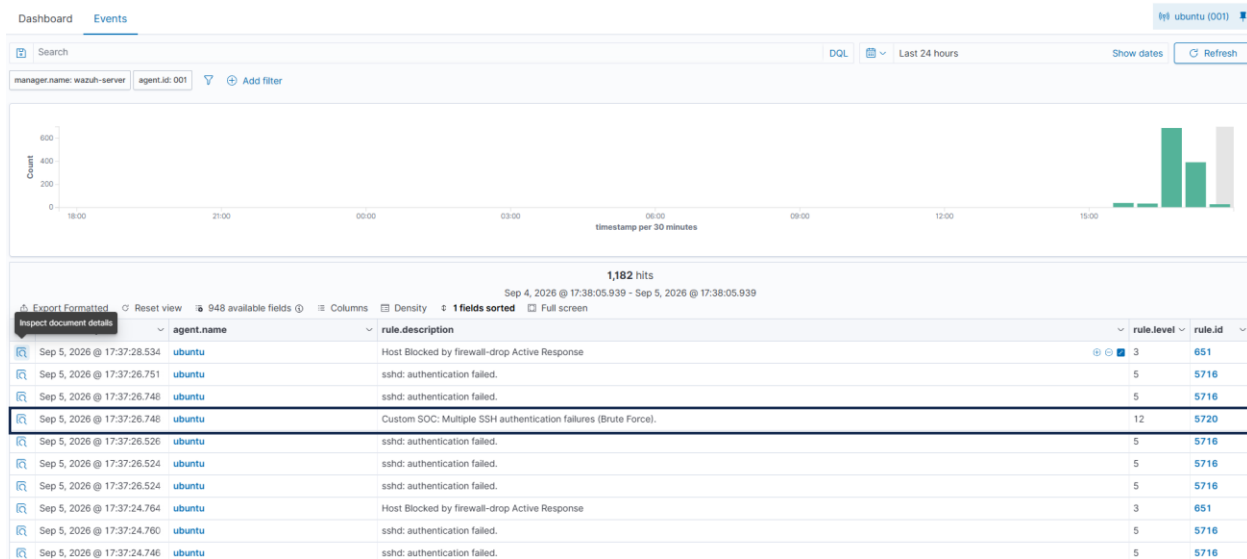
Not applicable for this specific incident (Direct payload delivery via external download).

2.5 Attack Execution

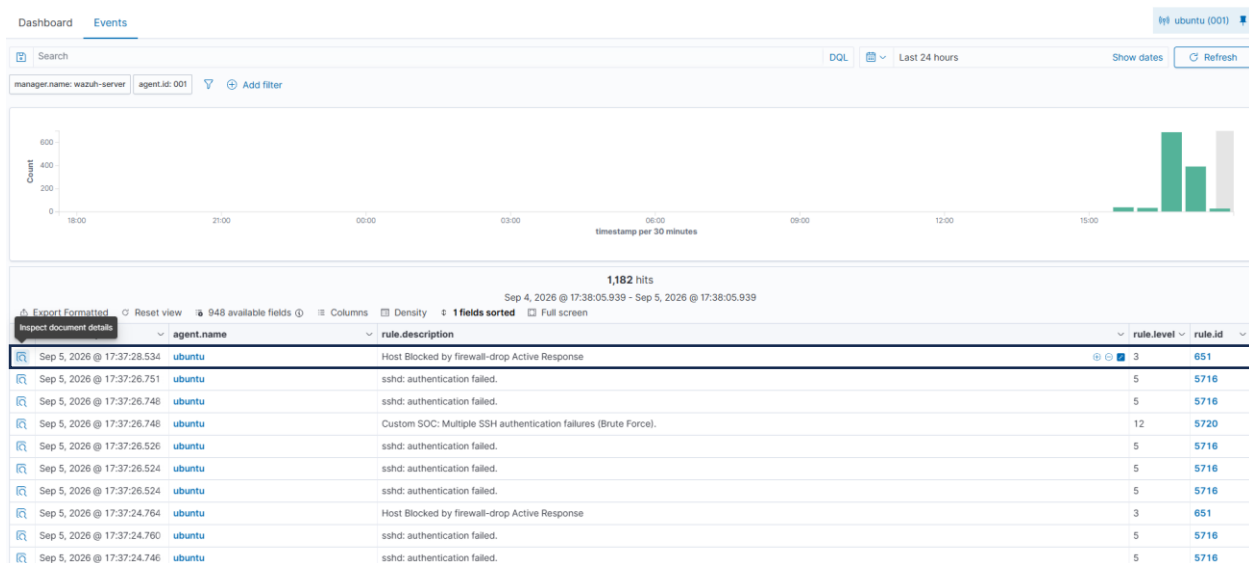
- An automated multi-threaded dictionary attack (hydra -t 4) was launched from the external Kali Linux node against the Ubuntu server's SSH daemon to harvest valid credentials.

3. Screenshot Evidence / Alerts (SIEM)

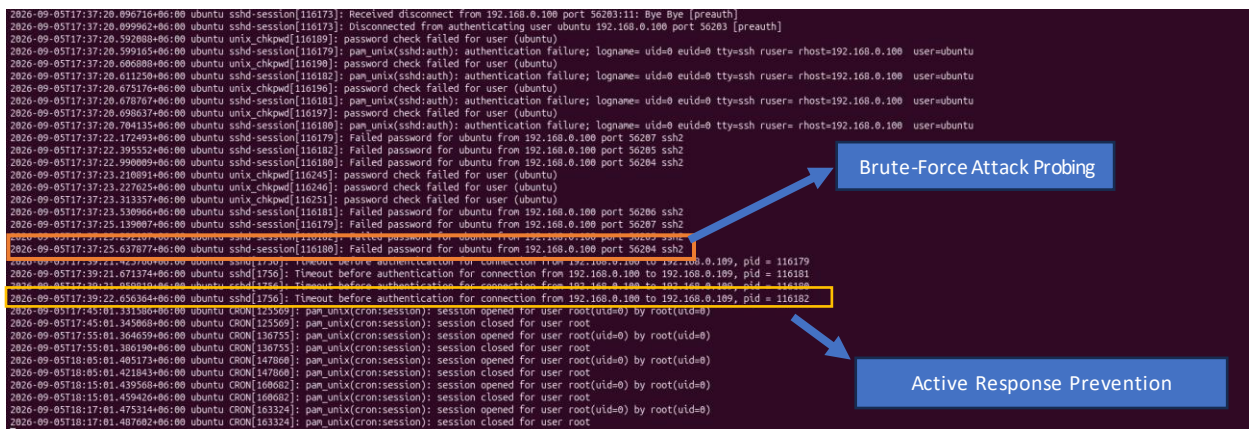
During the assault, the attacker pushed rapid login iterations, producing baseline failure events (5716). Upon crossing the configured 6-attempt threshold within 60 seconds, Wazuh successfully elevated the event priority.



- **Custom Threat Detection:** Rule ID 5720 successfully triggered at 17:37:24 with **Level 12 severity**, indicating a high-frequency brute-force attempt matching MITRE ATT&CK T1110.
- **Raw Telemetry Logs:** Continuous background authentication failures (Rule 5716, Level 5) were logged by the Ubuntu agent as connection threads attempted logins.
- **Automated Containment Verification:** Immediately upon rule escalation, **Rule ID 651** fired (Host Blocked by firewall-drop Active Response), confirming that the SIEM successfully instructed the host firewall to drop incoming malicious traffic.



- **Trigger Vector:** Linked directly to custom Rule 5720, bypassing the need for manual SOC analyst intervention.
- **Action Performed:** Dynamic execution of the local firewall-drop script on the target Ubuntu endpoint (192.168.0.109).
- **Enforced Isolation Window:** Applied a temporary network ban of **120 seconds** against the offending source IP (192.168.0.100), successfully severing active connection streams.



- **Credential Guessing Phase:** System logs captured sequential Failed password for ubuntu from 192.168.0.100 events alongside internal unix_chkpwd failures, verifying iterative password probing.
- **Connection Drop & Timeout Phase:** Following the active response deployment, subsequent log entries reflect Timeout before authentication and dropped connections (drop connection from 192.168.0.100), proving that the mitigation rules successfully blocked further packet processing.

4. Incident Ticketing

 Add epic /  KAN-98

 **Wazuh Alert: Custom SOC: Multiple SSH authentication failures (Brute Force).**

 **Description**

Incident Summary:
Agent: ubuntu
Severity Level: 12
Rule ID: 5720

Threat Details:
Custom SOC: Multiple SSH authentication failures (Brute Force).

Raw Telemetry Data:

```
1 {
2   "timestamp": "2026-09-05T11:37:26.748+0000",
3   "rule": {
4     "level": 12,
5     "description": "Custom SOC: Multiple SSH authentication failures (Brute Force).",
6     "id": "5720",
7     "mitre": {
8       "id": [
9         "T1110"
10      ],
11     "tactic": [
12       "Credential Access"
13     ],
14     "technique": [
15       "Brute Force"
16     ]
17   },
18 }
```

- **Incident Title / Summary:** Wazuh Alert: Custom SOC: Multiple SSH authentication failures (Brute Force).
- **Agent:** ubuntu
- **Severity Level:** 12

- **Rule ID:** 5720
- **MITRE ATT&CK Mapping:** Tactic Credential Access, Technique Brute Force (T1110)
- **Operational Note:** Automated REST API payload delivery successfully converted high-severity SIEM telemetry into a trackable Jira incident ticket (KAN-98).

5. Timeline of Detected Events

List detected events in chronological order.

Date / Time	Source IP	Destination IP	Event / Alert	Notes
05/09/2026 (17:37:24)	192.168.0.100	192.168.0.109	Rule 651: Host Blocked by Active Response	6th sequential failure within the 60-second timeframe window.
01/09/2026 (+1s)	192.168.0.100	192.168.0.109	Rule 651: Host Blocked by Active Response	Firewall rules updated dynamically to drop source traffic for 120s.

6. Source of Attack (IP Addresses & Methods Used)

Attacker Details	Target Details	Method / Tooling
192.168.0.100	Ubuntu Endpoint SSH service (192.168.0.109:22).	Dictionary Brute-Force

Note: Due to parallel thread utilization (-t 4), minor packet bleed occurred post-containment before complete network termination took effect, which is standard in high-velocity automated attacks.

7. Recommendations for Detection & Blocking

7.1 Immediate Containment

- The threat was successfully isolated and neutralized via automated firewall mitigation. No manual intervention or remediation is required for this event window...

7.2 Longer-Term Recommendations

- **Enforce Key-Based Authentication:** Disable password-based SSH authentication entirely across production environments, restricting access strictly to cryptographic SSH keys.
- **Refine Rate Limiting:** Implement secondary network-layer hardening (such as fail2ban or enhanced perimeter firewall throttling) to complement host-level SIEM controls.
- **Expand Threat Intel Correlation:** Feed recurrently malicious attack source IPs into perimeter blacklists to drop hostile traffic prior to reaching internal system boundaries.

— End of Report —